Changbo Wang

wang.changb@northeastern.edu | 929-417-6238 | Boston, MA
linkedin.com/in/wcb424 | github.com/wangcb424 | wangcb424.github.io/portfolio/

EDUCATION

Northeastern University **Boston, MA**
B.S. in Computer Science and Game Development | GPA: 3.68 Sep 2024 - May 2028
Relevant Coursework: Algorithms & Data; C++ Programming; Program Design and Implementation

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, JavaScript, TypeScript, SQL, HTML/CSS; OOP
Technologies: Spring Boot, Spring Data JPA, React, PostgreSQL, Unity, WordPress, Shopify
Tools & Testing: Git/GitHub, GitHub Actions, Docker Compose, JUnit, Playwright, Flyway, Figma

PROJECTS

CourseFlow - Course Seat Tracker & AI Assistant Java, Spring Boot, React, PostgreSQL | Live | GitHub

- Launched a React/TypeScript PWA using Northeastern public course data, with local watchlists, seat thresholds, and alerts while the page is visible.
- Trained a 116 KB bilingual TF-IDF intent classifier in Python and deployed it in-browser; grounded replies in validated course queries without external LLM calls.
- Built a separate Spring Boot/PostgreSQL edition with email-code login, session/CSRF protection, per-user watchlists, shared polling, and a transactional notification outbox with retries.
- Automated JUnit, proxy, and Playwright tests, including 31 assistant checks; validated isolation, retry handling, unpublished terms, and PostgreSQL integration in GitHub Actions.

Card Duel - Turn-Based Strategy Game C#, Unity | Play | GitHub

- Built a solo C#/Unity strategy game with a 6x6 exploration grid and turn-based combat; published a browser-playable build on Unity Play with a dedicated GitHub Pages game page.
- Composed card effects from ScriptableObjects; validated costs and targets before execution and tracked status activation and expiry across turns while separating content from mutable game state.
- Integrated an in-game LLM-powered enemy AI that evaluates combat state and player move patterns, then selects attacks based on component health; tested low-health and edge-case scenarios.
- Implemented JSON save/load with checksums, atomic file replacement, and state validation; preserved combat hands, random-generator state, and reward flags to prevent duplicate claims after reload.

The Garbage Man - 3D Top-Down Roguelike C#, Unity | GitHub | Play

- Developed and released a browser-playable 3D roguelike in Unity; composed enemies from interchangeable movement and weapon prefabs while reusing shared health and damage logic.
- Implemented a persistent run manager with PlayerPrefs checkpoints to restore level progression, health, currency, and upgrades across scene changes and subsequent play sessions.

Heartline - UX Case-Study Website JavaScript, HTML, CSS | Live | GitHub

- Published a responsive UX case-study site for a student friendship prototype, presenting user research, eight prototype screens, Figma embeds, and usability findings through GitHub Pages.

EXPERIENCE

Northeastern University / Zhongmiao **Boston, MA**
Frontend Engineer Oct 2026 - Present

- Develop responsive React/Vite features for a K-12 learning game, translating curriculum and gameplay requirements into browser-ready experiences; verify behavior with automated and browser-based QA.

Innospire Solutions Inc. **Queens, NY**
Information Technology Intern Sep 2023 - Dec 2023

- Supported requirements analysis and technical specifications for business applications; contributed to implementation, testing, and troubleshooting to address operational software needs.